
MediCare Pro

Smart Healthcare Management System — a full-stack platform for patients, doctors, and administrators.

Agenda

01 Foundations

- 01 Project Overview & Problem Statement
- 02 Objectives & Target Users
- 03 Main Features
- 04 Project Statistics
- 05 Tech Stack
- 06 System Architecture & Database

02 Deep Dive & Showcase

- 07 Security & Feature Breakdown
- 08 UI Showcase — All Modules
- 09 Demo Flow & Live Demo
- 10 Challenges & Solutions
- 11 Strengths & Future Enhancements
- 12 Team & Conclusion

WHAT IS MEDICARE PRO?

One platform. Every appointment, secured.

Connecting patients, doctors, and admins on one platform.

Problem Statement

Manual scheduling doesn't scale.

Clinics relying on phone bookings face conflicts, poor visibility, and no central oversight.

APPOINTMENT CONFLICTS

Phone-based scheduling leads to double-bookings and time clashes across clinics.

POOR COMMUNICATION

No direct digital channel between patients and doctors for updates or follow-ups.

NO DISCOVERY

Patients struggle to find verified specialists filtered by specialty and rating.

NO CENTRALIZATION

Scattered records make platform-wide monitoring and analytics impossible.

Project Objectives

- 01** Digitize Healthcare — move phone bookings online.
- 02** Improve Communication — direct doctor-patient contact.
- 03** Reduce Conflicts — prevent automatic double-bookings.
- 04** Centralize Management — one platform, three roles.
- 05** Responsive UX — consistent experience on every device.

Target Users

01 Patients & Doctors

Patients register, search and filter verified doctors, book and cancel appointments, pay securely, and leave reviews.

Doctors register for approval, manage their weekly schedule, accept or complete appointments, and maintain their professional profile.

02 Administrators

Administrators manage all users, approve doctor registrations, monitor every appointment platform-wide, and track revenue and system health through analytics dashboards.

Main Features

Six core capabilities, one platform.

Every feature is built around the three-role experience.

AUTHENTICATION

JWT tokens, bcrypt password hashing, protected routes, and role-based sessions.

DOCTOR DISCOVERY & BOOKING

Search by name or specialty, filter by rating, and book available time slots instantly.

SCHEDULE & PRACTICE MANAGEMENT

Doctors configure weekly availability and slot duration, and accept or complete appointments.

ANALYTICS & REVIEWS

Admin revenue dashboards, appointment breakdowns, and a star-rating review system.

Project Statistics



27

Total Pages

Patient, doctor, admin & public



34

REST API Routes

41 endpoints across 8 modules



7

Database Tables

SQL Server, fully relational



3

User Roles

Patient · Doctor · Admin



Stripe Integrated

Checkout, refunds & sync



JWT Secured

Stateless, role-based auth

MODERN, INDUSTRY-STANDARD TECHNOLOGIES

Tech Stack



React 19



Vite



Node.js



Express.js



SQL Server



Stripe



JWT



Git



GitHub

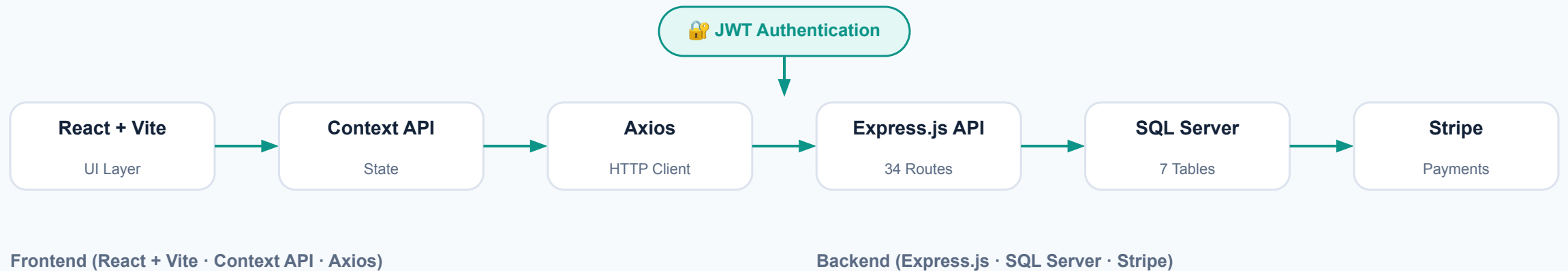


Axios

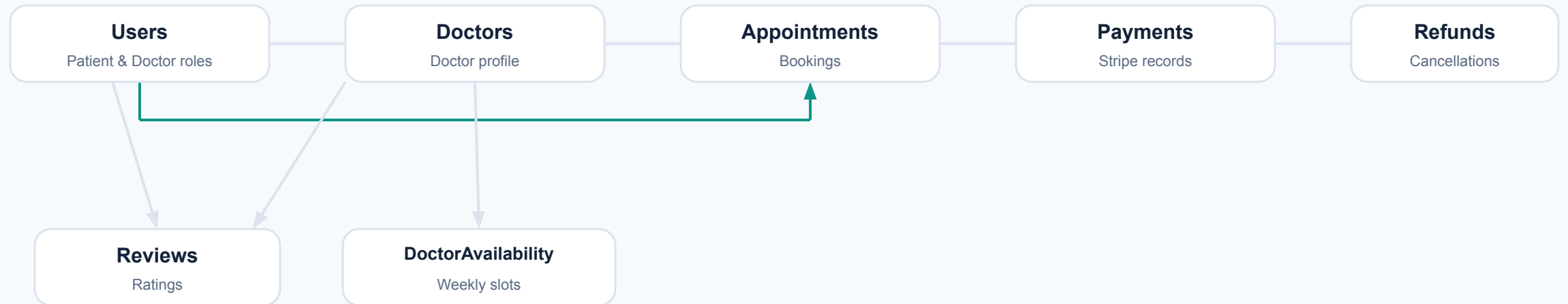
Also powered by React Context API for global state and Bootstrap 5 for responsive styling.

System Architecture

The React frontend calls the Express API over authenticated REST requests, which reads/writes SQL Server and coordinates Stripe payments.



Database Design



Patients and Doctors are both Users, distinguished by role. 7 tables · 6 views · 13 stored procedures · 2 triggers.

API Architecture

4

Authentication

Register, login, get & update profile

6

Doctor

List, view, update, create, delete

3

Patient / User

Profile & user statistics

6

Appointments

Book, list, update, cancel, delete

4

Availability

Slots, weekly schedule, updates

5

Payments

Stripe checkout, verify, refund, history

4

Reviews

Add, list, check, delete

9

Admin

Stats, revenue, approvals, management

Security

Security built in, not bolted on.

Every layer of MediCare Pro enforces authentication and data protection.

JWT AUTHENTICATION

Stateless, signed tokens with role claims secure every session.

PASSWORD HASHING

bcrypt with salt rounds — passwords are never stored in plain text.

PROTECTED ROUTES & RBAC

Frontend route guards and backend middleware isolate Patient, Doctor, and Admin access.

SECURE STRIPE PAYMENTS

Checkout sessions and idempotent verification keep every transaction safe.

Features by User Role

One codebase, three purpose-built experiences.

Feature	Patient	Doctor	Admin
Doctor Discovery & Appointments	✓ Search, book & cancel	✓ Accept & complete	✓ Monitor all
Payments & Refunds	✓ Pay & view history	–	✓ Manual refunds
Scheduling & Availability	✓ View open slots	✓ Set weekly hours	–
Reviews & Verification	✓ Leave reviews	✓ Get verified	✓ Approve doctors
Analytics & User Management	–	–	✓ Dashboards & users



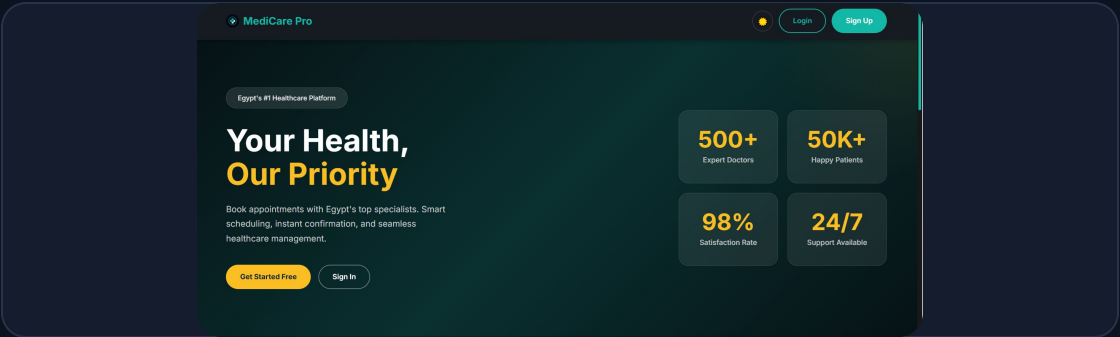
MediCare Pro logo

BRAND IDENTITY

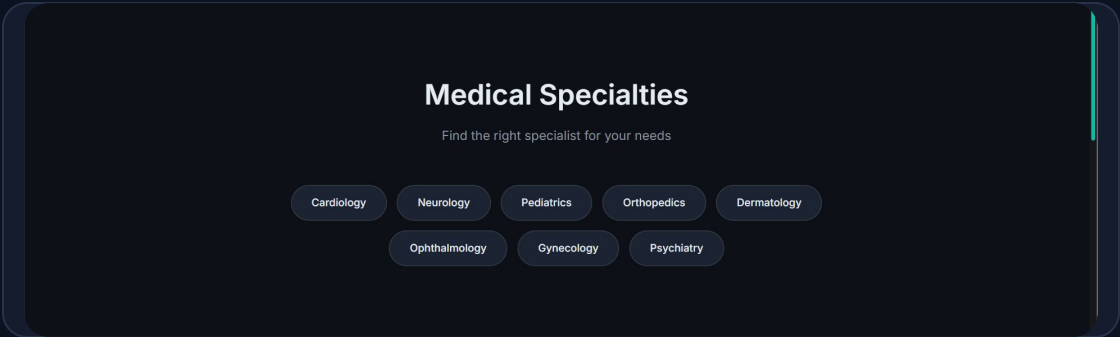
A Trusted Digital Health Mark

The MediCare Pro mark pairs a medical cross with a heartbeat pulse — a simple symbol of care and vitality. A dark, teal-accented interface carries that identity across every patient, doctor, and admin experience.

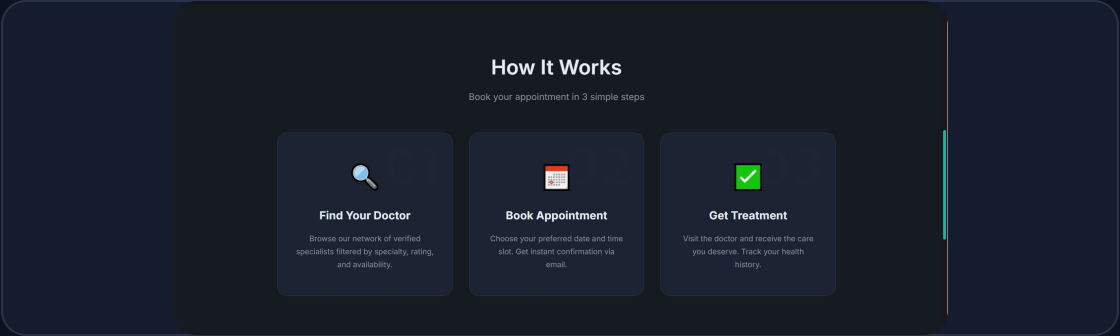
UI Showcase — Landing Page



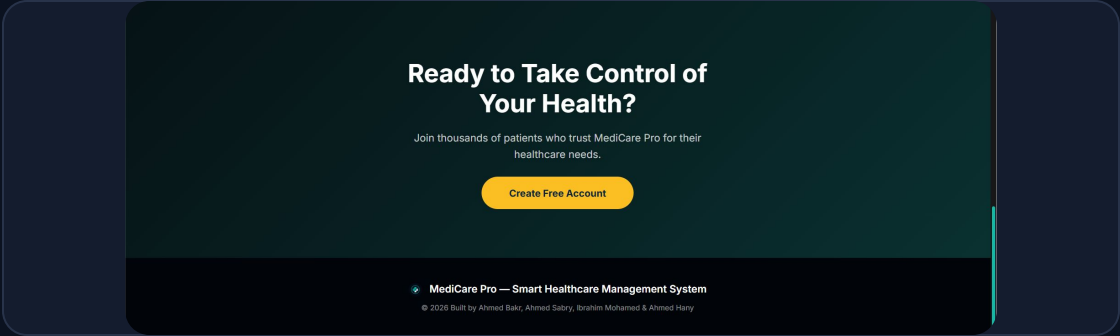
Hero Section



Medical Specialties

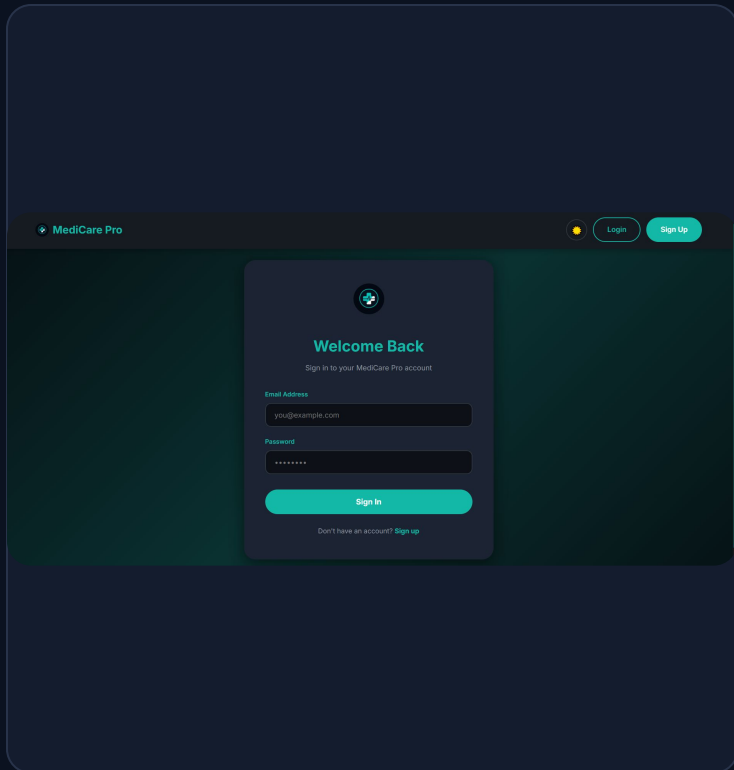


How It Works



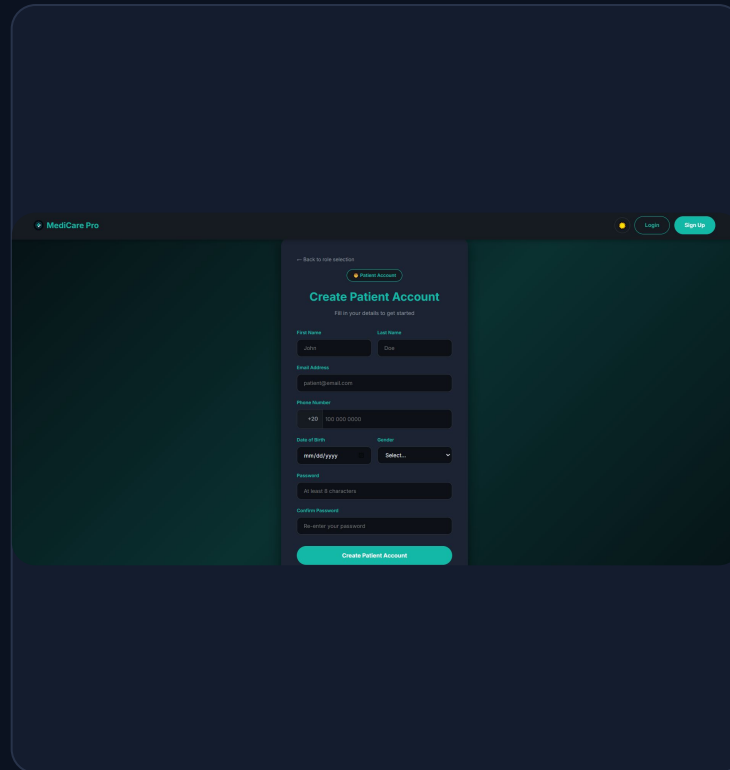
Call to Action & Footer

UI Showcase — Authentication



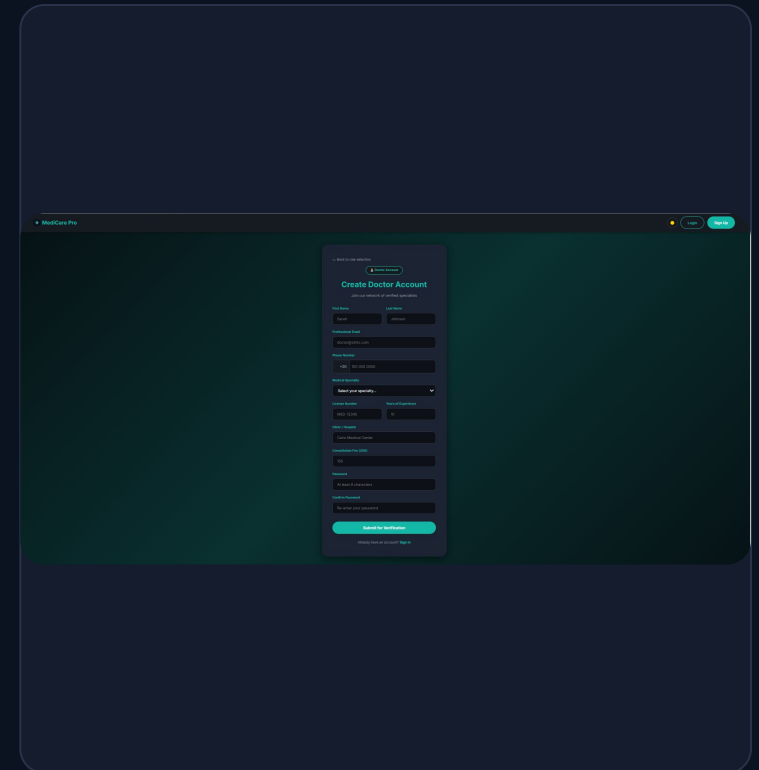
The login screen features a dark blue header with the 'MediCare Pro' logo, a moon icon, and 'Login' and 'Sign Up' buttons. The main content area has a dark teal background. A white card in the center contains a medical cross icon, the text 'Welcome Back', and a sub-header 'Sign in to your MediCare Pro account'. Below this are input fields for 'Email Address' (with 'you@example.com' as a placeholder) and 'Password' (with masked characters). A large blue 'Sign In' button is at the bottom, with a link 'Don't have an account? Sign up' underneath.

Login



The 'Create Patient Account' screen has a dark blue header with the 'MediCare Pro' logo, a moon icon, and 'Login' and 'Sign Up' buttons. The main content area has a dark teal background. A white card contains a back arrow, a 'Patient Account' tab, and the title 'Create Patient Account' with the subtitle 'Fill in your details to get started'. The form includes fields for 'First Name' and 'Last Name', 'Email Address', 'Phone Number', 'Date of Birth', 'Gender', 'Password', and 'Confirm Password'. A blue 'Create Patient Account' button is at the bottom.

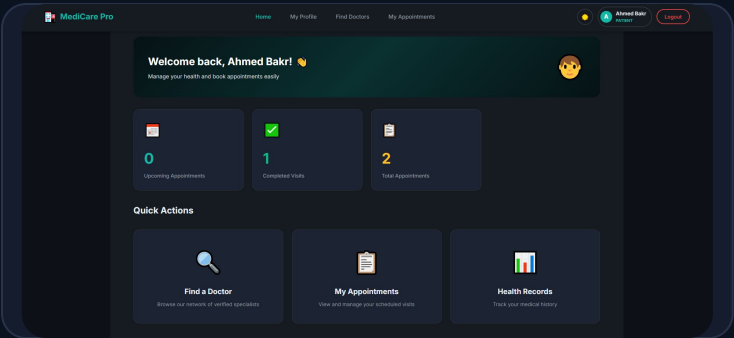
Patient Sign Up



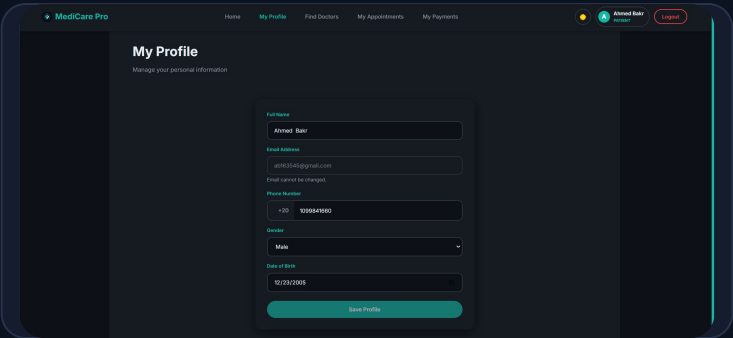
The 'Create Doctor Account' screen has a dark blue header with the 'MediCare Pro' logo, a moon icon, and 'Login' and 'Sign Up' buttons. The main content area has a dark teal background. A white card contains a back arrow, a 'Doctor Account' tab, and the title 'Create Doctor Account' with the subtitle 'Get the details of your account'. The form includes fields for 'First Name', 'Last Name', 'Email Address', 'Phone Number', 'Date of Birth', 'Gender', 'Specialty', 'License Number', 'Years of Experience', 'Current Location', and 'Current Address'. A blue 'Create Doctor Account' button is at the bottom.

Doctor Sign Up

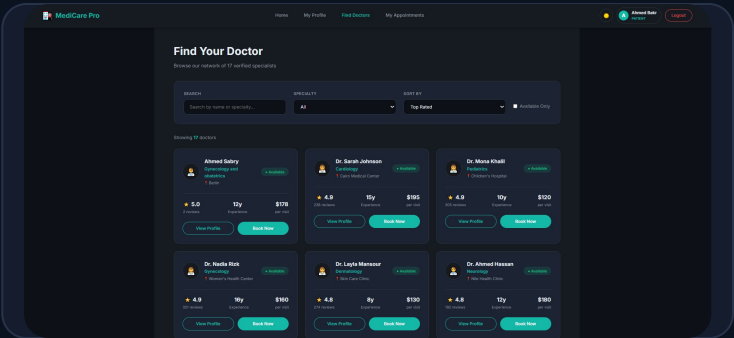
UI Showcase — Patient Module



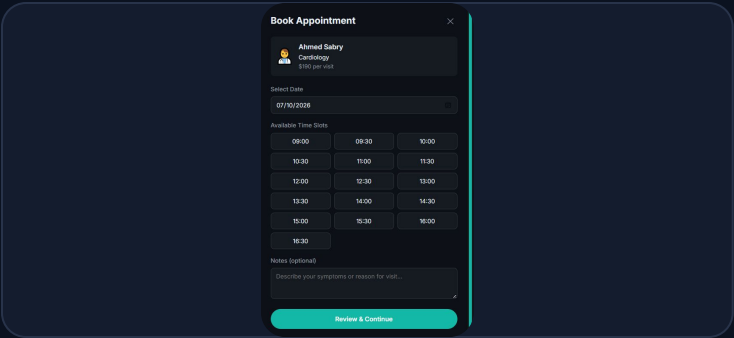
Patient Home



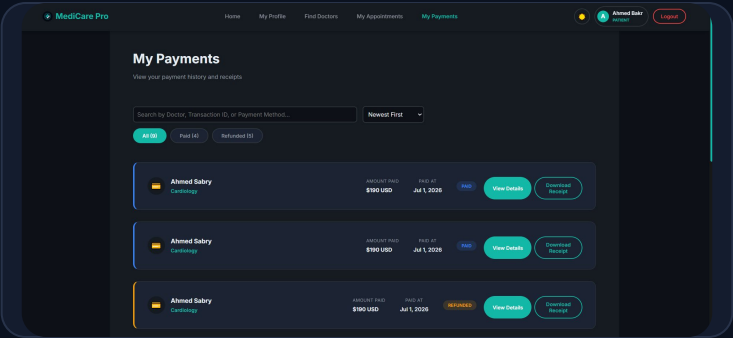
My Profile



Find Doctors

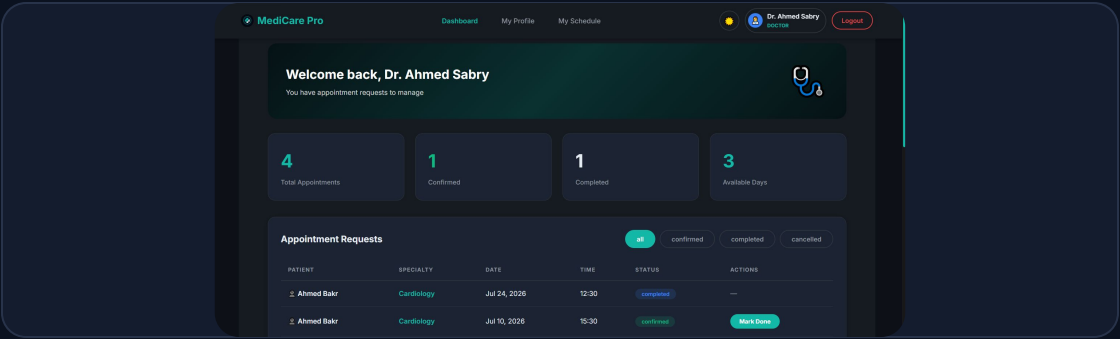


Book Appointment

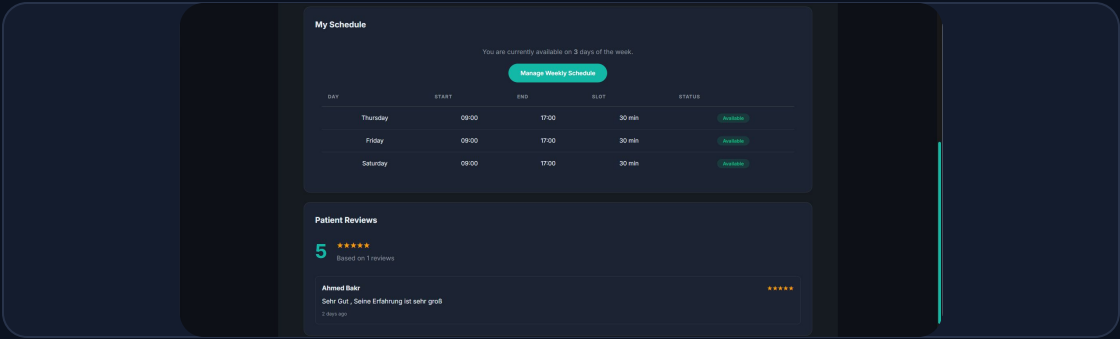


My Payments

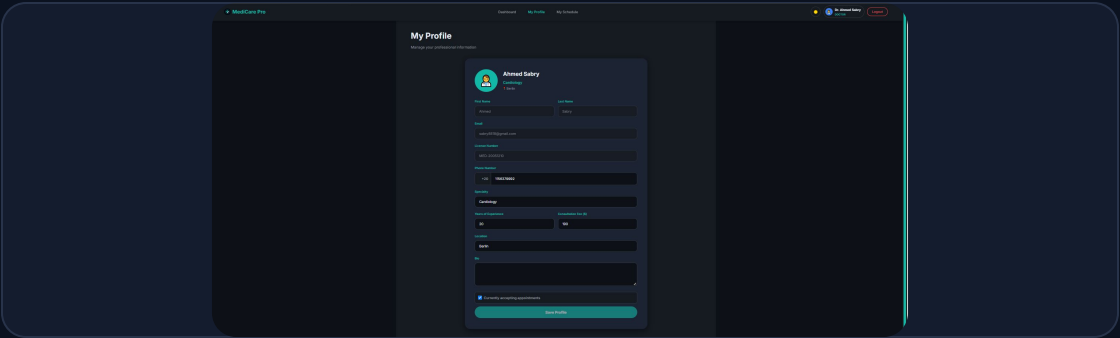
UI Showcase — Doctor Module



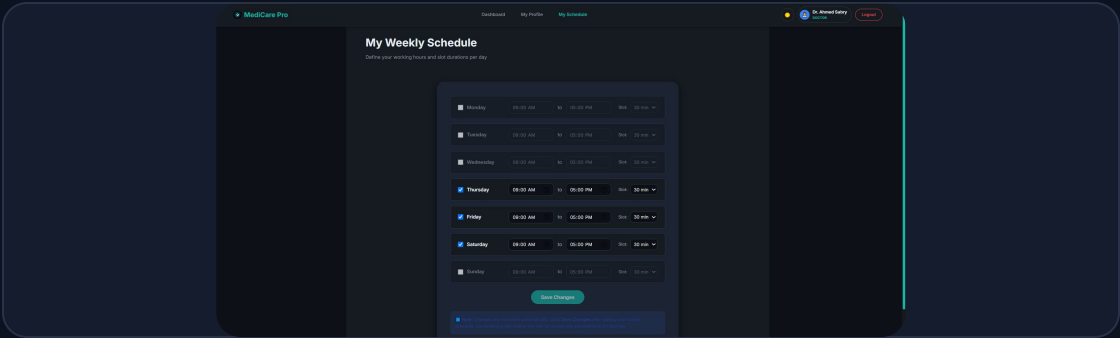
Dashboard — Requests



Schedule & Reviews

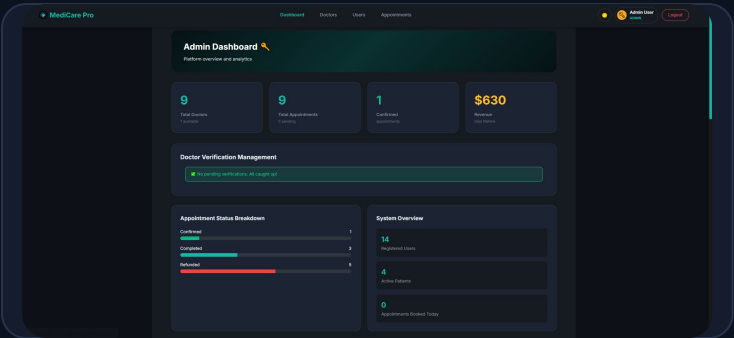


Professional Profile

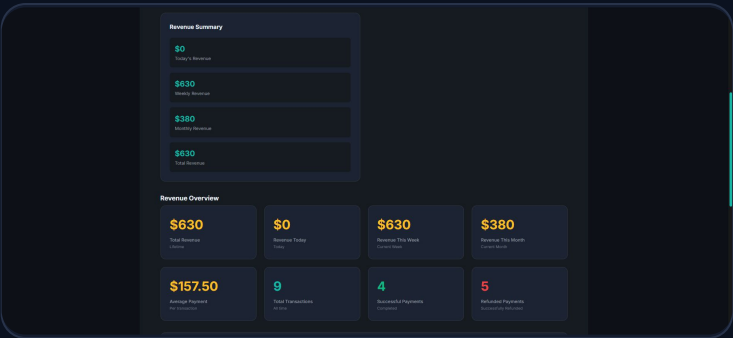


Weekly Schedule Editor

UI Showcase — Admin Panel



Admin Overview



Revenue Summary

Manage Doctors

View and manage all registered doctors

All Doctors (9)

Search by name, email, or specialty...

DOCTOR NAME	EMAIL	SPECIALTY	EXPERIENCE	FEE	STATUS	VERIFICATION	ACTIONS
Dr. Sarah Johnson	sarah@depi.com	Cardiology	23y	\$200	Available	Approved	Edit
Dr. Ahmed Hassan	ahmed@depi.com	Neurology	12y	\$180	Available	Approved	Edit
Dr. Omar Farouk	omar@depi.com	Orthopedics	18y	\$185	Busy/Offline	Approved	Edit
Dr. Layla Mansour	layla@depi.com	Dermatology	8y	\$130	Available	Approved	Edit
Dr. Karim Nabil	karim@depi.com	Ophthalmology	14y	\$170	Available	Approved	Edit

Manage Doctors

Manage Users

View and manage all platform users

All Users (14)

Search by name or email...

Active Patients Doctors Admins

USER NAME	EMAIL	ROLE	JOINED	STATUS	ACTIONS
Admin User	admin@depi.com	Admin	2028-05-12	Active	Current Account
Dr. Sarah Johnson	sarah@depi.com	Doctor	2028-05-12	Active	Delete
Dr. Ahmed Hassan	ahmed@depi.com	Doctor	2028-05-12	Active	Delete
Dr. Omar Farouk	omar@depi.com	Doctor	2028-05-12	Active	Delete
Dr. Layla Mansour	layla@depi.com	Doctor	2028-05-12	Active	Delete

Manage Users

All Appointments

Monitor and manage all appointments across the platform

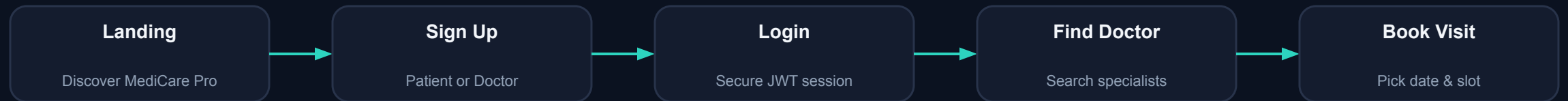
All (9) Confirmed (3) Completed (3) Cancelled (3)

Search patient, doctor, or specialty...

PATIENT	DOCTOR	SPECIALTY	DATE	TIME	FEE	STATUS	ACTIONS
Ahmed Bakr	Ahmed Satory	Cardiology	7/24/2026	12:30	\$180	Completed	Delete
Ahmed Bakr	Ahmed Satory	Cardiology	7/10/2026	15:30	\$180	Confirmed	Delete
Ahmed Bakr	Ahmed Satory	Cardiology	7/10/2026	15:00	\$180	Cancelled	Delete
Ahmed Bakr	Fady Mina	Pediatrics	7/4/2026	16:00	\$125	Completed	Delete

All Appointments

Demo Flow — Getting Started



Demo Flow — Booking to Review



Challenges & Solutions

CHALLENGE

Time Slot Collisions

Preventing double-bookings across concurrent requests.

Payment State Sync

Users could drop off mid-payment.

Orphaned Records

Deleting a user could break linked data.

Complex State Management

Three roles needed to share auth state.



SOLUTION

✓ DB transactions validate every slot before it is booked.

✓ Stripe Session IDs confirm payment before booking is created.

✓ Soft deletes protect data integrity across every table.

✓ Context API & Axios interceptors centralize auth state.

Project Strengths

01 Secure by design — JWT, bcrypt, and RBAC from day one.

02 A true multi-role platform with three dashboards.

03 Smart scheduling plus a live revenue analytics dashboard.

04 A modern stack — React, Vite, Node.js, and SQL Server.

Future Enhancements

01 Coming Next

Doctor–Patient Chat — real-time messaging between doctors and patients.

Video Consultation — real-time teleconsultation for remote visits.

Notifications — email & SMS alerts for appointment reminders.

02 On the Roadmap

AI Doctor Matching — smart doctor suggestions based on symptoms & history.

Medical Records — digital health records and prescription management.

Mobile Application — native iOS & Android apps for on-the-go access.

Team Contributions

Eight specialists, one integrated healthcare platform.

Eight specialists, one integrated healthcare platform. Each team member owned a specific system module, working together to deliver a complete end-to-end healthcare solution.

AHMED BAKR

Project Lead & System Architecture

Project planning, system architecture, deployment, documentation, presentation, tech stack, application integration, and overall project coordination.

AHMED HANY

Backend Core

Express.js server, database connection, middleware, business logic, error handling, and backend architecture.

IBRAHIM MOHAMED

Admin Module

Admin dashboard, analytics, revenue tracking, doctor management, user management, appointments, and

AHMED SABRY

Frontend Core

React architecture, routing, reusable components, UI/UX, theme management, Axios management, Axios integration, responsive design, and navigation.

Team Contributions

Every module was designed to integrate seamlessly with the

Every module was designed to integrate seamlessly with the others, resulting in a scalable, secure, and maintainable healthcare platform.

MINA SABIR

Authentication & Security

Authentication, registration, JWT, protected routes, role-based authorization, and session management.

BASANT ALI

Patient Module

Patient dashboard, doctor search, booking journey, profile management, reviews, and patient experience.

MAHMOUD AMR

Doctor Module

Doctor dashboard, weekly schedule, availability management, doctor profile, and appointment handling.

FATIMA MOHAMED

Appointment & Payment

Appointment workflow, Stripe integration, payment processing, refund workflow, and appointment business rules.

Conclusion

01 Platform Capabilities

- ✓ Multi-Role Healthcare Platform
- ✓ Secure Authentication
- ✓ Role-Based Access Control
- ✓ Appointment Conflict Prevention

02 Business & Experience Wins

- ✓ Stripe Payment Integration
- ✓ Automatic Refund System
- ✓ Smart Appointment Scheduling
- ✓ Analytics Dashboard
- ✓ Responsive UI

Live Demo

A single continuous journey across all three roles, demonstrated live end to end.



MEDICARE PRO — SMART HEALTHCARE MANAGEMENT SYSTEM



Thank You

DEPI Graduation Project · 2026

github.com/A200B5/smart-healthcare-system

Ahmed Bakr · Ahmed Sabry · Ibrahim Mohamed · Ahmed Hany

Scan for GitHub repo

